



Soo Kyung Ahn

Software Engineer

✉ sahnsookyung@gmail.com 🔗 sookyungahn.com

🐙 github.com/sahnsookyung

🌐 https://www.linkedin.com/in/soo-kyung-ahn-9a7aa012a/

SUMMARY

Fullstack Software Engineer with 3.5 years' professional experience building reliable, high-throughput services for banking and digital goods platforms. Skilled in Kubernetes and AWS with strong observability, production readiness, and stakeholder management. Hands-on integration of LLM-powered features into backend workflows across local and GPU-hosted inference.

PROFESSIONAL EXPERIENCE

2022 – 2025
Digital Goods and
Services Team | Cape
Town, South Africa

Electrum Payments

Software Engineer

Built and operated services processing millions of transactions per day for top South African banks and retailers, improving transaction success rates and reducing incident MTTR via targeted observability dashboards.

- Led the development and maintenance of backend services for digital goods (airtime, vouchers, money transfers, lotto), powering high-volume transactions across South Africa's top banks and national retailers.
- Designed modular, plugin-based components enabling faster feature delivery and safer rollouts across multiple clients on the same platform.
- Drove test automation (JUnit, Mockito), code quality (+80% on SonarQube), and CI/CD robustness (GitHub Actions, CircleCI) to support frequent, low-risk production releases.
- Operated services on AWS EKS with Istio, owning deployment strategies, routing, and runtime diagnostics, and reducing incident MTTR through targeted Kibana/Elasticsearch dashboards.
- Partnered with DevOps to integrate Kafka, SQS/SNS, Route53, and observability tooling into developer workflows, improving deployment reliability and debugging speed.
- Prototyped internal AI-powered tools including an LLM-based code review bot and automated ticket triage system to streamline engineering and support workflows.

EDUCATION

2018 – 2020

B.Sc. Computer Science & Applied Statistics (Distinction)

University of Cape Town

- Capstone Project: Visualization environment for complex molecules in the domain of drug design.
- Dean's Merit List

2021

B.Sc. Honours in Computer Science (First Class GPA)

University of Cape Town

- Thesis Focus: Nature simulation studying various techniques for improving immersion and user engagement using Virtual Reality (Unity) & HCI.

TECHNICAL SKILLS

AI & ML

- LLM Orchestration
- Local and Remote GPU based inference (vLLM)
- Voice Pipelines
- Vector Databases, Embeddings, Semantic Search
- AI-native development

Infrastructure

- AWS(EKS, EC2, S3, Lambda, RDS, Route53, SQS/SNS)
- Kubernetes/Docker
- Observability Stack (ELK, Prometheus, Grafana)
- CI/CD (Github Actions, CircleCI)
- Redis

Frontend Engineering

- TypeScript
- React.js
- Web Components
- TailwindCSS
- Three.js
- Vite

Backend Engineering

- Python
- Java
- MySQL/PostgreSQL
- FastAPI
- REST API/WebSockets
- Event-Driven Architecture
- Playwright/Web scraping

Interactive Applications

- Unity 3D (C#)
- VR Development

AI ENGINEERING PROJECTS

JobScout

AI-Powered Job Matching Pipeline

- Full-stack intelligent system that automatically scrapes job listings from multiple platforms, extracts structured data using local LLMs (Ollama), and matches opportunities to user resumes using semantic vector search (pgvector).
- Key Features:
 - ETL pipeline with concurrent job ingestion, extraction, facet analysis, and embedding generation
 - Semantic matching engine using vector similarity between resume skills and job requirements
 - Fit/want scoring system to evaluate job alignment
 - Web dashboard (FastAPI + React) for browsing matches, viewing stats, and configuring alerts
 - Async notification system with Redis queue for Discord/email digests
 - Pipeline orchestration with preemption logic for concurrent execution
- Tech Stack: Python, FastAPI, React, PostgreSQL (pgvector), Redis, Ollama, Docker, SQLAlchemy

FRONTEND PROJECTS

Portfolio/Personal Website (sookyungahn.com)

<https://github.com/sahnsookyung/jiko-shoukaisho-website>

- An interactive single-screen portfolio website featuring a hand-drawn SVG landscape with clickable regions that trigger content overlays.
- Key Features:
 - Interactive SVG landscape with dynamic overlays and responsive tooltips
 - Multiple purpose-built content viewers (resume, photo gallery, timeline, thoughts)
 - Real-time Three.js particle system simulating gravitational lensing and "waterfall" effect from Korean words
 - Native Web Components architecture for modular, framework-less components
 - DOMPurify for security when sanitizing dynamically injected content
- Tech Stack: TypeScript, HTML5, Tailwind CSS v4, Three.js, Web Components, Vite